to receive and transmit a signal. The Tag replies the information from its memory bank. The reader will transmit to read the result to RFID computer program.

Two types of RFID tags were Active RFID and Passive RFID systems.

1. In a passive RFID tag, the power is supplied by the reader when radio waves from the reader are encountered by a passive RFID tag, the coiled antenna forms a magnetic field.
2. Battery powered RFID tag is installed with small battery that powers the broadcast of information

Main Components of a RFID System

- **A RFID tag:** It has silicon microchip attached to a small antenna and mounted on a substrate. See Figure 11.7

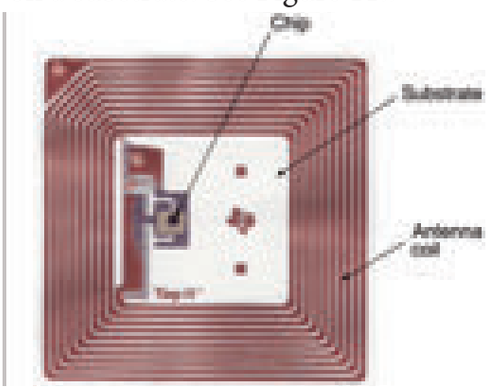


Figure 11.7 RFID Tag

- **A reader:** It has a scanner with antennas to transmit and receive signals, used for communication. See Figure 11.8



Figure 11.8 An RFID Reader

- **A Controller**: It is the host computer with a Microprocessor which receives the reader input and process the data.

Two types of RFID Systems:

1. **Active RFID system:** The tag has its own power source. These systems are used for larger distances and to track high value goods like vehicles.
2. **Passive RFID system:** The tag gets power from a reader antenna to the tag antenna. They are used for shorter range transmission.

11.2 Reference Model

11.2.1 OSI Model

Open System Interconnection (OSI) model was found in the year 1934, general framework that enables network protocols along with software and systems to be developed based on general set of guidelines. It describes the standards for the inter-computer communication. See Figure 11.9

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TIPS

There are many prompts used to remember the OSI layer order:

- Everyone Needs Data Processing.
- Everyone Should Try New Diet Pepsi.

OSI Layers:

1. **Physical Layer:** This is the 1st layer, it defines the electrical and physical specifications for devices.
2. **Data Link Layer:** It is the 2nd layer and it guarantees that the data transmitted

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